



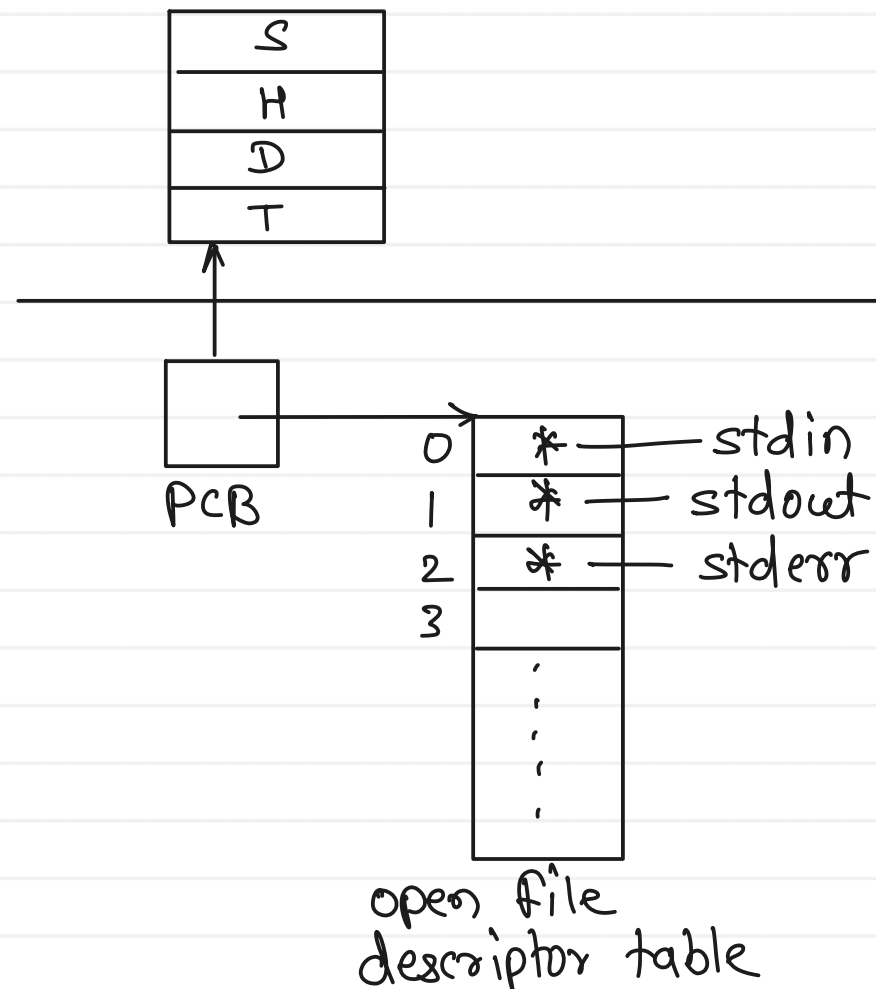
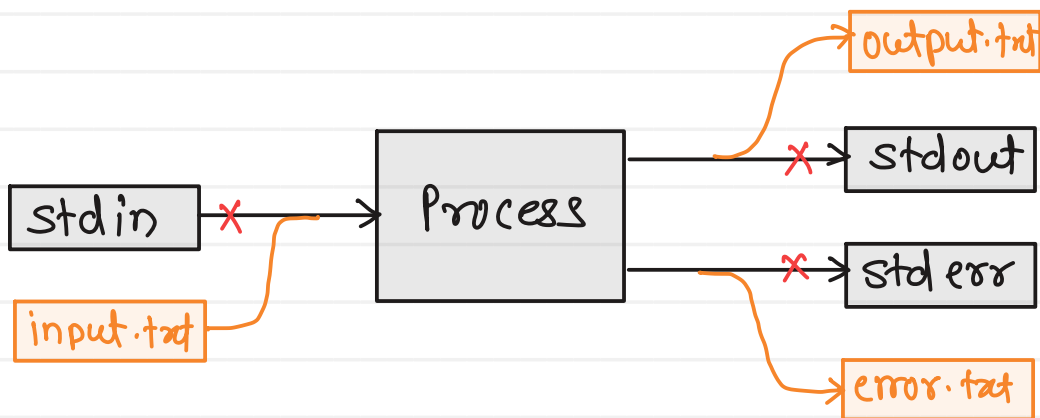
Sunbeam Institute of Information Technology
Pune and Karad

Module - Operating System Concepts and Linux Programming

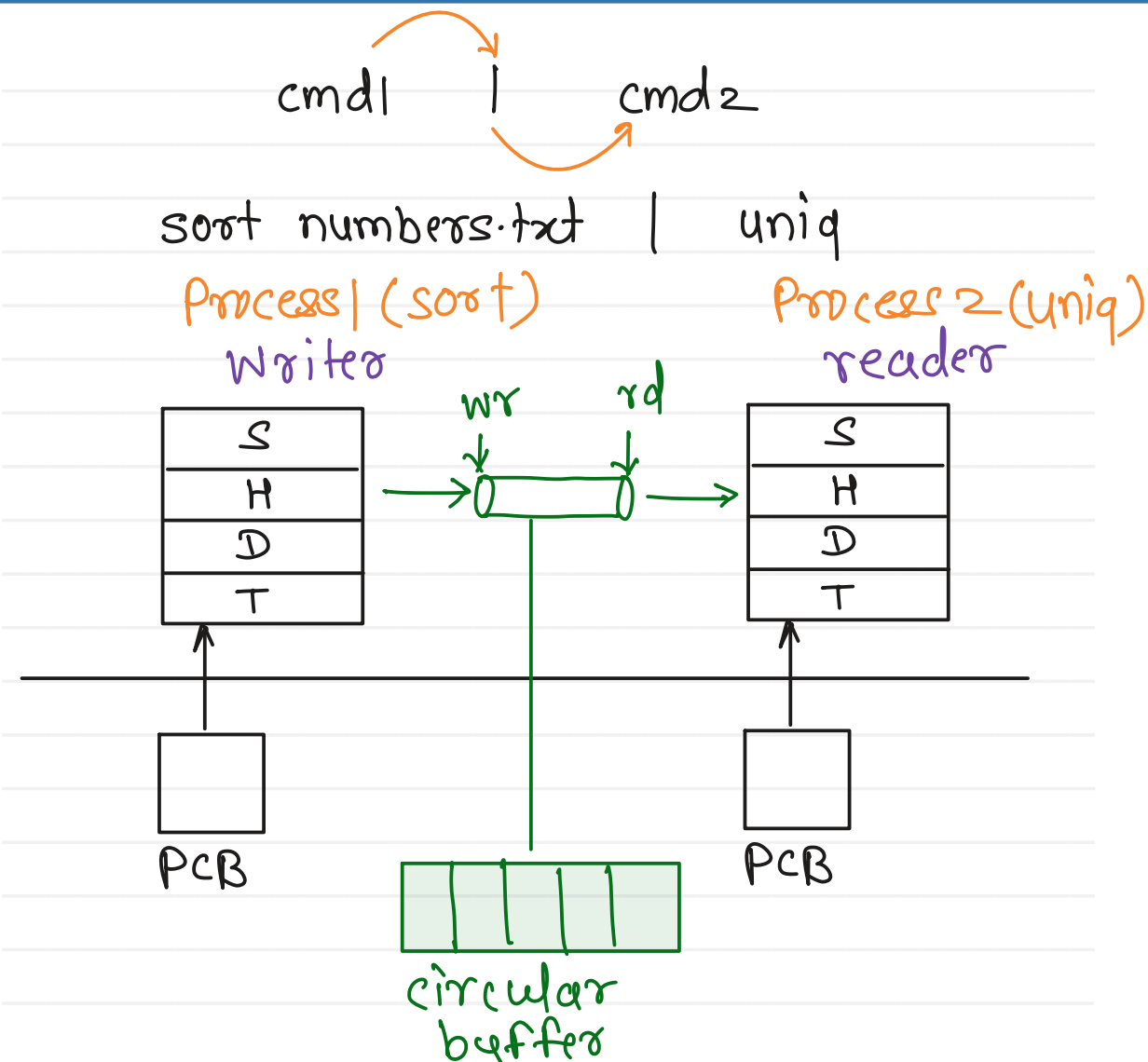
Trainer - Devendra Dhande

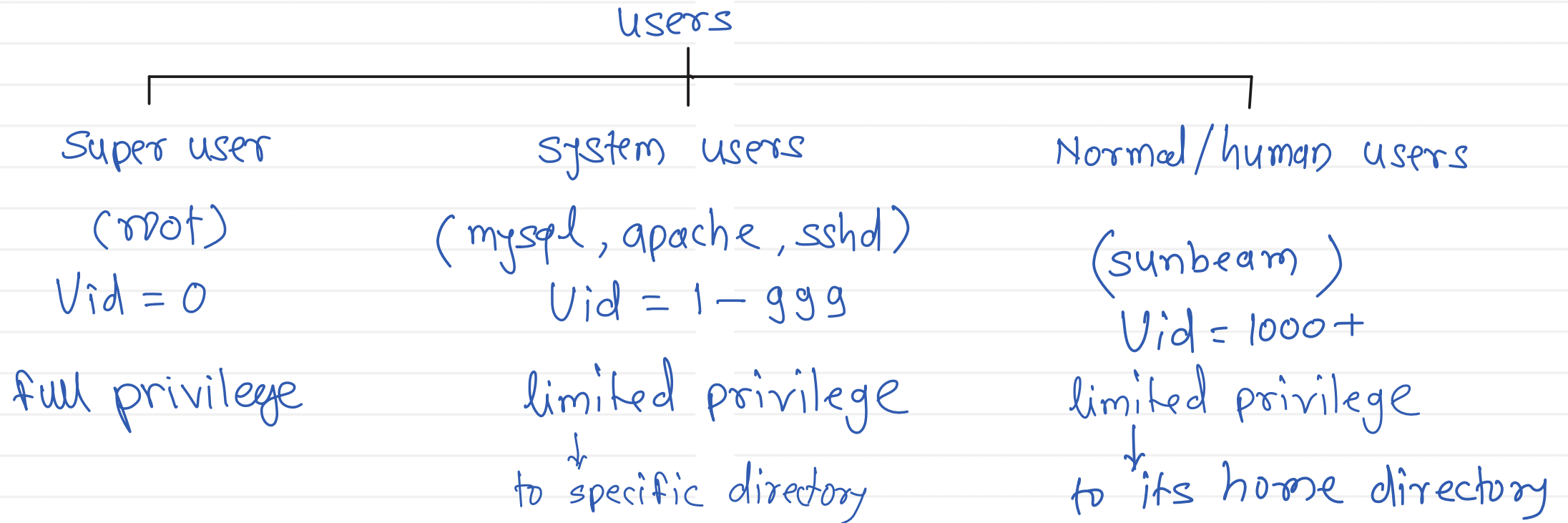
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- By default, standard streams (files) are opened for every process
 - stdin
 - stdout
 - stderr
- through these files only input/output will be done in a program



- pipe is an IPC mechanism used to communicate betⁿ two processes.
- it is a unidirectional IPC
- every pipe has two ends
 1. write
 2. read





File - types, permissions, groups

type → **-** rwXrwxr-x **1** sunbeam sunbeam 16753 Mar 29 05:49 sample.txt
 permissions/mode/access user/owner group timestamp

Types of file :

1. Regular file (-)
2. Directory file (d)
3. Link file (l)
4. Pipe file (p)
5. Socket file (s)
6. Char special file (c)
7. Block special file (b)

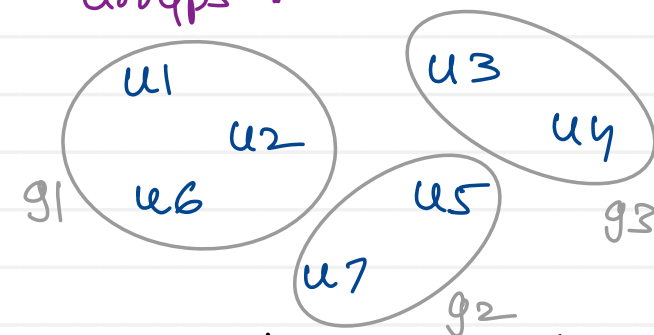
permissions :

r - read
 w - write
 x - execute

levels :

rwX rwX r-x
 ↑ ↑ ↑
 user/ group others
 owner

Groups :



u1 has created sample.txt
 Name: sample.txt

use/owner: u1 (rwX)
 group : g1 (u2, u6) (rwX)
 others : g2 & g3 (u3, u4, u5, u7) (r-x)

Types of commands :

1. Internal commands

- they are implemented within shell it self.

e.g. cd, alias, unalias

2. External commands

- they are implemented as seperate program.

e.g. cp, mv, ls

Types of Execution :

1. Synchronous execution :

- shell will wait for the completion of command.

e.g. \$ command

2. Asynchronous execution :

- shell will not wait for the completion of command.

e.g. \$ command &